Neural Networks and Deep Learning

Neural Networks: The Basics

A neural network is a model made of layers of simple units called neurons, loosely inspired by biological brains. Each neuron computes a weighted sum of its inputs, adds a bias, and passes the result through a non-linear activation function. Stacking many neurons lets the network represent complex relationships that a single linear model cannot.

A typical feedforward network has an input layer, one or more hidden layers, and an output layer. The values that the network learns are the weights and biases; training is the process of adjusting them so the network's outputs match the desired targets.

Layers and Activation Functions

Activation functions give a neural network its ability to model non-linear patterns. Without them, stacking layers would be equivalent to a single linear transformation. The rectified linear unit, or ReLU, outputs zero for negative inputs and the input itself for positive inputs, and it is the most common choice in hidden layers because it is simple and trains quickly.

Other activations include the sigmoid, which squashes values into the range zero to one and is useful for probabilities, and softmax, which turns a vector of scores into a probability distribution across classes at the output layer.

Training with Backpropagation

Neural networks are trained with backpropagation combined with gradient descent. First, a forward pass computes the network's prediction and a loss function measures how far that prediction is from the target. Backpropagation then applies the chain rule of calculus to compute the gradient of the loss with respect to every weight in the network.

Gradient descent uses those gradients to nudge each weight in the direction that reduces the loss, scaled by a learning rate. Repeating this forward-and-backward cycle over many batches of data gradually lowers the loss. Choosing a good learning rate matters: too large and training diverges, too small and it crawls.

Deep Learning Applications

Deep learning refers to neural networks with many hidden layers, which can learn hierarchical features directly from raw data. Convolutional neural networks excel at images by detecting edges, then shapes, then objects. Recurrent networks and transformers handle sequences such as text and speech.

These architectures power everyday applications including image recognition, machine translation, speech-to-text, and recommendation systems. Their main costs are the need for large datasets and significant compute for training, though a trained model can often run efficiently afterward.